Status	Finished
Started	Thursday, 20 November 2025, 12:00 PM
Completed	Thursday, 20 November 2025, 12:21 PM
Duration	20 mins 35 secs
Marks	3.00/3.00
Grade	**3.00** out of 3.00 (**100**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Bob is using the RSA encryption scheme with public key equal to $(n, x) = (3233, 1969)$.

a) Bob's RSA key is weak. Break his encryption and find his Bob's private key:

$y =$ 2689

Your last answer was interpreted as follows:

$$2689$$

It might help to know that the following is a list of the first prime numbers:

$$[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79]$$

b) Alice uses Bob's public key to encrypt a message, which she then sends over a public channel.

The encrypted message is

$$M = 1220$$

Find the unencrypted (plain text) message

$m =$ 1220

Your last answer was interpreted as follows:

$$1220$$

The following calculator can be used for computing exponents $\mod (n)$:

Base:
Exponent:
Modulus:
Result:

Question **2**

Correct

Mark 1.00 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

$$[101010, 011001, 001001, 100100]\,.$$

$l_{min} =$ 1

Your last answer was interpreted as follows:

$$1$$

$l_{max} =$ 5

Your last answer was interpreted as follows:

$$5$$

Question **3**

Correct

Mark 1.00 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \to (\mathbb{Z}/2)^{\times 5}$$

be the encoding function given by $E(w_1 w_2 w_3) = w_1 w_2 w_3 w_4 w_5$, where

$$\begin{bmatrix} w_4 = w_3 + w_2 + w_1 \\ w_5 = w_1 \end{bmatrix}$$

a) Find the generator matrix G for the encoding E

$G =$

1 0 0 1 1
0 1 0 1 0

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

b) Find the parity check matrix H for the encoding E.

$H =$

1 1 1 1 0
1 0 0 0 1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

c) Let $r = 01110$ be a received message. Compute the syndrome of r with respect to the encoding E.

Syndrome of r = 10

Your last answer was interpreted as follows:

$$10$$

Syntax guide: Give your answers to **a)** and **b)** as matrices. Use a space character to separate entries in the same row, and line break to separate rows.

Syntax guide: Give your answer to **c)** as a bitstring. E.g. if you think the answer to a question is "1011", then you should enter **1011** .

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 389

Signing code: 14357

Your last answer was interpreted as follows:

$$14357$$

◄ Test 3 (topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod(n) and Lagrange's theorem)

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